

Group Consolidated Requirements Worksheet Table

#	Requirement (as written)	Source (role/stakeholders)	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes/Resolution
1	The system shall store enough client info such that, in the case an item is damaged or lost, we may follow-up with the client over email	Shop Owner	FAIL - Conflict with requirement 20	FAIL	PASS	FAIL (item instead of gear)	PASS	PASS	Resolution: Replace with two new requirements: “The system shall identify customers using their name and phone number only, without the need to create an account or password.” “The system shall demand a deposit from customers when checking out certain high-value items.” See Requirement 20 for details.
2	The system shall store client identifying info securely (i.e. guard against cyberattacks)	Shop Owner	PASS	FAIL	FAIL	PASS	FAIL	PASS	Resolution: rewrite as “The system shall satisfy PCI-DSS (Payment Card Industry Data Security Standard)”
3	System shall validate client info, so clients cannot give fake information	Shop Owner	PASS	FAIL	FAIL	PASS	PASS	PASS	Resolution: rewrite as “The system shall use OTP (One Time Passwords) to verify client identity”
4	The system shall be easily extendable for future developers	Shop Owner	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: rewrite to “The system shall be easily extendable for future development, such as online shopping”
5	The system shall allow us to check the status of gear being rented	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
6	The system shall track metrics on rented gear and create easy-to-understand graphs and visuals using these metrics	Shop Owner	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: Split into 2 new requirements: 1. “The system shall record which customer currently has each rented equipment.” 2. “The system shall provide a report on what gear is out right now, what’s overdue, and which items get rented most”
7	The system shall have easy to read guides on how to start the system, stop the system, and troubleshoot errors in the system	Shop Owner	PASS	FAIL	PASS	PASS	FAIL	PASS	Resolution: Rewrite as “The system shall have guides targeted at nontechnical users for how to start the system, stop the system, and troubleshoot errors in the system”
8	The system shall allow employees to edit gear info in case employees input the wrong information	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
9	The system shall allow us to toggle which gear/items need deposits and which don’t. This includes manual toggle, as well as applying filters (e.g. all items that cost more than \$200, all items that are larger than 1m^3)	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	

10	The system shall cost no more than \$10 a month to run	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
11	The system shall be fast and simple to use	Rental Clerk	PASS	FAIL	FAIL	PASS	FAIL	PASS	Resolution: split it into several requirements: “The system shall take no longer than half a second to verify my credentials, take no longer than half a second to load each page, and take no longer than half a second to reflect any updates I input as an employee.” “The system shall require no more than 15 minutes of onboarding to be used effectively by any employee”
12	The system shall know who previously just used a piece of gear	Rental Clerk	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: rewrite as “The system shall track who previously rented each piece of gear”
13	The system shall be able to search by the name of the client or gear	Rental Clerk	PASS	FAIL	FAIL	PASS	PASS	PASS	Resolution: Split requirements more clearly “The system shall be able to search up gear items by name as well as the name of rental clients” “The system shall add an additional short numerical identifier for all customers, in case multiple clients have the exact same name.”
14	The system shall be able to flag gear that has been returned damaged as “out of service”	Rental Clerk	PASS	PASS	PASS	PASS	PASS	PASS	
15	The system shall utilise an ID and deposit-based system when clients take out high value gear to advise clients to be more careful (and to partly pay for damages for high value gear if there are any)	Rental Clerk	FAIL - Conflict with requirement 19	PASS	PASS	PASS	PASS	PASS	Resolution: Replace with two new requirements: “The system shall identify customers using their name and phone number only, without the need to create an account or password.” “The system shall demand a deposit from customers when checking out certain high-value items.” See Requirement 20 for details.
16	The system shall display to clients whether gear they want to rent is available in real time	Customer	PASS	PASS	PASS	PASS	PASS	PASS	
17	The system shall allow clients to check gear out within several seconds or less	Customer	PASS	PASS	PASS	PASS	PASS	PASS	
18	The system shall display to clients what gear they are due to return and when to return them in an easy-to-read and intuitive fashion	Customer	PASS	PASS	PASS	PASS	PASS	PASS	
19	The system shall not use deposits for gear	Customer	FAIL - Conflict with requirements 1 & 15	PASS	PASS	PASS	PASS	PASS	Resolution: Replace with two new requirements: “The system shall identify customers using their name and phone number only, without the need to create an account or password.”

									“The system shall demand a deposit from customers when checking out certain high-value items.” See Requirement 20 for details.
20	The system shall not ask more from me other than name and phone number when renting equipment	Customer	FAIL - Conflict with requirement 1 & 15	PASS	PASS	PASS	PASS	PASS	Resolution: based on client needs of: 1. Quick checkout of items w/o need for account 2. Avoiding need to give personal info (as giving too much feels like client has been accused) and store needs of: 1. Accountability on lost/damaged items 2. Tracking metrics on who is renting what gear we reached the following compromise: “The system shall identify customers using their name and phone number only, without the need to create an account or password.” “The system shall demand a deposit from customers when checking out certain high-value items.” This ensures clients can check out gear w/o feeling accused, while also satisfying accountability requirements and need for metrics for the store employees. We replace requirements 1, 15, 19, and 20 with this requirement.
21	The system shall allow me to easily extend my rentals	Customer	PASS	FAIL	FAIL	PASS	FAIL	PASS	Resolution: rewrite as “The system shall allow clients to extend their rentals for an additional 3 days or further for additional charges in a visually clear and responsive manner (if the rented gear is available for rent)
22	The system shall keep clients informed of upcoming due dates for rented gear	Customer	PASS	PASS	PASS	PASS	PASS	PASS	
23	The system shall be online during store times so that clients may return their gear and have the return be properly logged	Customer	PASS	PASS	PASS	PASS	PASS	PASS	